

Aprendizaje de Máquina

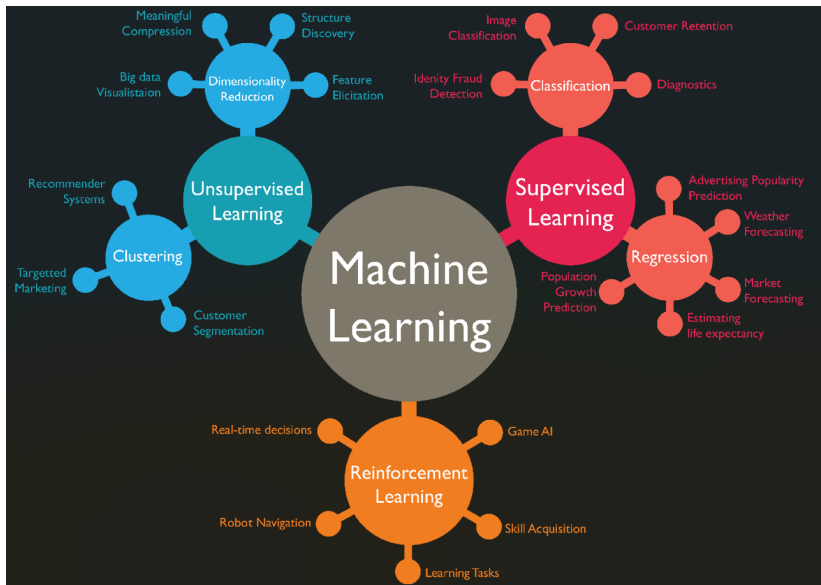
U1: Introducción

Héctor Maravillo

Section 1

Basic Concepts

Machine Learning



Clustering

Clustering is an **unsupervised** machine learning algorithm that organizes and classifies different objects, data points, or observations into groups or clusters based on **similarities** or patterns.

Some applications:

- Image segmentation
- Market segmentation
- Anomaly detection

Similarities

Quantifying the term **similar** and **dissimilar** depends very much on the type of clusters involved.

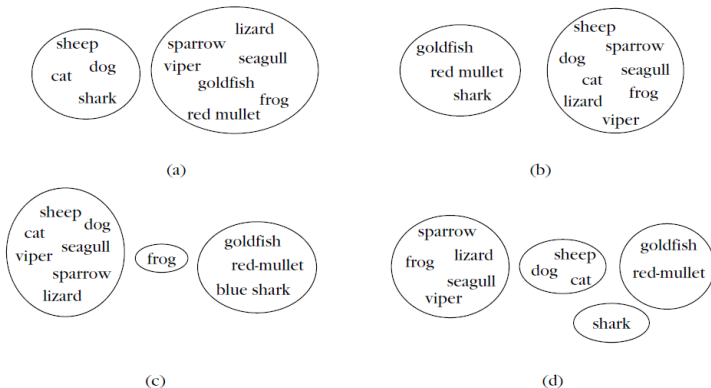


Figure: Clustering criterion: a) Reproductive Methods. b) Lung Presence. c) Habitats. d) Reproductive Methods and Lung Presence

Clustering task

The basic steps to develop a clustering task are the following:

- **Feature selection.** Features should maximize relevant information while minimizing redundancy
- **Proximity measure.** This measure quantifies how *similar* or *dissimilar* two feature vectors are.
- **Clustering criterion.** The clustering criterion may be expressed via a cost function or some other types of rules.
- **Clustering algorithms.**
- **Validation of the results.** Once the results of the clustering algorithm have been obtained, we have to verify their correctness.
- **Interpretation of the results.** In many cases, the expert in the application field must integrate the results of clustering with other experimental evidence and analysis in order to draw the right conclusions.

m-Clustering

Let X a data set, usually over an l -dimensional space,

$$X = \{x_1, \dots, x_N\}$$

We define $\mathcal{P}$ as an ***m*-clustering** if it is the **partition** of X into m sets (**clusters**), $C_1, \dots, C_m$, so that the following three conditions are met:

$$C_i \neq \emptyset, \quad i = 1, \dots, m \quad (1)$$

$$\bigcup_{i=1}^m C_i = X \quad (2)$$

$$C_i \cap C_j = \emptyset, \quad i \neq j, i, j = 1, \dots, m \quad (3)$$

In addition, the vectors contained in a cluster C_i are **more similar** to each other and **less similar** to the feature vectors for the other clusters.

Fuzzy clustering

A **fuzzy clustering** of X into m clusters is characterized by m functions u_j where

$$u_j : X \rightarrow [0, 1], \quad j = 1, \dots, m \quad (4)$$

$$\sum_{j=1}^m u_j(x_i) = 1, \quad i = 1, \dots, N \quad (5)$$

$$0 < \sum_{i=1}^N u_j(x_i) < N, \quad j = 1, \dots, m \quad (6)$$

These are called **membership function**. Each vector x belongs to more than one cluster simultaneously **up to some degree**, which is quantified by the corresponding value of u_j in the interval $[0, 1]$.

Number of Possible Clustering

Let $S(N, m)$ denote the number of all possible clusterings of N vectors into m groups, which is called **Stirling number**

$$S(N, m) = \frac{1}{m!} \sum_{i=0}^m (-1)^{m-i} \binom{m}{i} i^N$$

This number is only valid for the case in which the number of clusters is fixed. Especially for $m = 2$, becomes

$$S(N, 2) = 2^{N-1} - 1$$

Example:

$$S(15, 3) = 2,375,101$$

$$S(25, 8) = 690,223,721,118,368,580$$

$$S(100, 5) \approx 10^{68}$$

Section 2

Proximity Measures

Dissimilarity

Definition

A **dissimilarity measure (DM)** d on X is a function

$$d : X \times X \rightarrow \mathbb{R}$$

where X is a data set, such that,

$$\exists d_0 \in \mathbb{R} : \quad -\infty < d_0 \leq d(x, y) < \infty, \quad \forall x, y \in X \quad (7)$$

$$d(x, x) = d_0, \quad \forall x \in X \quad (8)$$

$$d(x, y) = d(y, x), \quad \forall x, y \in X \quad (9)$$

Dissimilarity

Definition

A **metric DM** is a **dissimilarity measure** that also satisfies

$$d(x, y) = d_0 \quad \Leftrightarrow \quad x = y \quad (10)$$

$$d(x, z) \leq d(x, y) + d(y, z), \quad \forall x, y, z \in X \quad (11)$$

Similarity

Definition

A **similarity measure (SM)** s on X is defined as

$$s : X \times X \rightarrow \mathbb{R}$$

such that,

$$\exists s_0 \in \mathbb{R} : \quad -\infty < s(x, y) \leq s_0 < \infty, \quad \forall x, y \in X \quad (12)$$

$$s(x, x) = s_0, \quad \forall x \in X \quad (13)$$

$$s(x, y) = s(y, x), \quad \forall x, y \in X \quad (14)$$

Similarity

Definition

A **metric SM** is a similarity measure that also satisfies

$$s(x, y) = s_0 \quad \Leftrightarrow \quad x = y \quad (15)$$

$$s(x, y)s(y, z) \leq [s(x, y) + s(y, z)] s(x, z) \quad \forall x, y, z \in X \quad (16)$$

DM for real-valued vectors

- The **weighted l_p metric DMs** or **Minkowski distance**

$$d_p(x, y) = \left(\sum_{i=1}^l w_i |x_i - y_i|^p \right)^{\frac{1}{p}}$$

where x_i, y_i are the i th coordinates of vectors $x, y \in X$, $i = 1, \dots, l$, and $w_i \geq 0$ is the i th weight coefficient.

- **Invariant** to any translation and rotation only for $p = 2$ (Euclidean distance).
- Features with large values and variances tend to **dominate** over other features.
- The Manhattan, Euclidean and Chebyshev distance are special cases of the Minkowski distance.

- The **Manhattan norm**, l_1 norm or **city-block taxicab) distance**

$$d_1(x, y) = \sum_{i=1}^I |x_i - y_i|$$

- The **Euclidean distance** or l_2 norm,

$$d_2(x, y) = \sqrt{\sum_{i=1}^I (x_i - y_i)^2}$$

- The l_∞ norm or **Chebyshev distance**,

$$d_\infty(x, y) = \max_{1 \leq i \leq I} |x_i - y_i|$$

It can be shown that

$$d_\infty(x, y) \leq d_2(x, y) \leq d_1(x, y)$$

● Mahalanobis distance

$$d(x, y) = \sqrt{(x - y)^T \Sigma^{-1} (x - y)}$$

where Σ is the covariance matrix of X .

Properties:

- **Invariant** to any non-singular linear transformation.
- Tend to form **hyper-ellipsoidal clusters**.
- The data do not need to be scaled or standardized before calculating the distance.
- Calculating the Mahalanobis distance can be **computationally expensive** due to the need to invert the covariance matrix.
- If Σ is the identity, the Mahalanobis distance reduces to the Euclidean distance.

- **Canberra distance**

$$d(x, y) = \begin{cases} 0 & \text{for } x = y = 0 \\ \sum_{i=1}^I \frac{|x_i - y_i|}{|x_i| + |y_i|} & \text{for } x \neq 0 \text{ or } y \neq 0 \end{cases}$$

- Very sensitive to small.
- Useful when relative changes in small values are important.

SM for real-valued vectors

- **Cosine similarity measure** or **angular separation**

$$s_{\text{cosine}}(x, y) = \frac{x^T y}{||x|| ||y||}$$

where

$$||x|| = \sqrt{\sum_{i=1}^I x_i^2}, \quad ||y|| = \sqrt{\sum_{i=1}^I y_i^2}$$

are the lengths of the vectors x and y , respectively.

- The most commonly used distance in **document area**.
- This measure is **invariant** to rotations but not linear transformations.
- **Ignore** magnitude of vectors.
- Not useful for high degree of correlation data

- **Pearson's correlation coefficient**

$$r_{\text{Pearson}}(x, y) = \frac{x_d^T y_d}{||x_d|| ||y_d||}$$

where $x_d = (x_1 - \bar{x}, \dots, x_l - \bar{x})^T$ and $y_d = (y_1 - \bar{y}, \dots, y_l - \bar{y})^T$ with x_i, y_i being the i th coordinates of x and y , respectively, and

$$\bar{x} = \frac{1}{l} \sum_{i=1}^l x_i, \quad \bar{y} = \frac{1}{l} \sum_{i=1}^l y_i$$

- Pearson's correlation takes values between -1 and 1 .
- Unable to detect the magnitude of differences of two variables.

Contingency table

Let x a vector whose coordinates belong to the finite set $F = \{0, 1, \dots, k-1\}$, where k is a positive integer. So, there are exactly k^l vectors $x \in \mathbb{F}^l$.

One can imagine these vectors as vertices in an l -**dimensional grid**.

Consider $x, y \in \mathbb{F}^l$ and let

$$A(x, y) = [a_{ij}], \quad i, j = 0, 1, \dots, k-1$$

be a $k \times k$ matrix, where the element a_{ij} is the number of places where the first vector has the i symbol and the corresponding element of the second vector has the j symbol, with $i, j \in \mathbb{F}$. This matrix is also known as a **contingency table**.

Example: if $l = 6$, $k = 3$ and

$$x = (0, 1, 2, 1, 2, 1)^T$$

$$y = (1, 0, 2, 1, 0, 1)^T$$

then the matrix $A(x, y)$ is equal to

Example: if $l = 6$, $k = 3$ and

$$x = (0, 1, 2, 1, 2, 1)^T$$

$$y = (1, 0, 2, 1, 0, 1)^T$$

then the matrix $A(x, y)$ is equal to

$$A(x, y) = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Also,

$$\sum_{i=0}^{k-1} \sum_{j=0}^{k-1} a_{ij} = l$$

DM for discrete-valued vectors

- **Hamming distance.** It is defined as the **number of places where two vectors differ**.

Using the matrix X , the Hamming distance $d_H(x, y)$ is

$$d_H(x, y) = \sum_{i=0}^{k-1} \sum_{j=0, j \neq i}^{k-1} a_{ij}$$

that is, the summation of all the off-diagonal elements of A , which indicate the position where x and y differ. Only for same length vectors or strings.

- The l_1 **distance**. For discrete-valued vectors is also possible use the l_1 distance, defined as in the continuous case:

$$d_1(x, y) = \sum_{i=1}^l |x_i - y_i|$$

When the vectors are binary-valued, the l_1 and the Hamming distance coincide.

SM for discrete-valued vectors

- **Tanimoto measure**¹. It is inspired by the comparison of sets. If X and Y are two sets.

The Tanimoto measure between sets is:

$$\frac{|X \cap Y|}{|X| + |Y| - |X \cap Y|} = \frac{|X \cap Y|}{|X \cup Y|}$$

The Tanimoto measure between two sets is the **ratio** to the number of elements they have in **common** to the number of all different elements.

¹Also, called **Jaccard similarity**.

DM for discrete-valued vectors

If we have ordinal features and interpret the value of the i th coordinate of a vector as the degree to which the vector possesses the i th feature. According to this interpretation, the pairs $(x_i, y_i) = (0, 0)$ are less important than the others.

Then, the **Tanimoto measure** is defined as

$$s_T(x, y) = \frac{\sum_{i=1}^{k-1} a_{ii}}{\sum_{i=1}^{k-1} \sum_{j=0}^{k-1} a_{ij} + \sum_{i=0}^{k-1} \sum_{j=1}^{k-1} a_{ij} - \sum_{i=1}^{k-1} \sum_{j=1}^{k-1} a_{ij}}$$
$$s_T(x, y) = \frac{\sum_{i=1}^{k-1} a_{ii}}{n_x + n_y - \sum_{i=1}^{k-1} \sum_{j=1}^{k-1} a_{ij}}$$

DM for discrete-valued vectors

The figure shows a 3x3 grid of coordinate pairs. A solid black border encloses the right two columns, specifically the pairs (0,1), (0,2), (1,1), (1,2), (2,1), and (2,2). A dashed black border encloses the left two columns, specifically the pairs (0,0), (1,0), (1,1), (1,2), (2,0), and (2,1). The intersection of these two regions is the middle column containing the pairs (0,1), (1,1), and (2,1).

$(0, 0)$	$(0, 1)$	$(0, 2)$
$(1, 0)$	$(1, 1)$	$(1, 2)$
$(2, 0)$	$(2, 1)$	$(2, 2)$

Figure: The elements of a contingency table taken into account for the computation of the Tanimoto measure.

References

- Theodoridis, S. & Koutroumbas, K. *Pattern Recognition*, 4th ed., Academic Press, 2009
- Xu, R. & Wunsch, D. Survey of Clustering Algorithms, *IEEE Transactions on Neural Networks*, 2005.